# Chronus Interview Experience

## ONLINE TEST:

The online test was conducted in the HackerEarth platform where 4 coding questions were asked and we had to complete within 1.25 hours (1 hour 15 minutes). You can switch to any questions during the test.

## Coding questions:

1. There are n cars with arriving and departure times are given. These cars are arriving to a parking lot where Rs. 50 is collected for each car. Only one car can be parked at a time. You have to find the maximum amount that can be collected.
   This question is little similar to https://leetcode.com/problems/non-overlapping-intervals/

2. There are n buildings with no. of floors specified in an array. There are k builders. We have to allot the buildings to builders such that they construct only consecutive buildings. Each builder takes one month to complete one floor. We have a budget of (Rs.) b initially. We spend (Rs.) 5 in each month. You have to find maximum profit amount (budget b – amount spent).

   E.g.: Consider there are 5 buildings with floors [5,10,35,10,15], there are 3 builders and we have an initial budget of Rs. 200.

   If we split as Builder 1 – 5,10; Builder 2 – 35; Builder 3 – 10,15; we will have a max. month of 35 so profit = 200 – 5*35 = 25.

3. There was a music concert planned with given reserved seats in form of coordinates as (x,y). There are speakers which covers only 90 degrees around it (in any one of the quadrants). We can place speakers at any of these reserved seats. Your task is to find minimum number of speakers needed such that it covers all the audience (reserved seats).

   E.g.: The coordinates are (-2, -2), (-1, -1), (0, 0), (1, 1), (2, 2). Placing 1 speaker (-2, -2) can cover all remaining coordinates.

4. On a particular day, an employee can be 'A', 'L', 'P' where 'A' – Absent, 'L' – Late, 'P' – Present. An employee can be rewarded with a bonus only if he is
   Only 'A' for two days.
   Not 'L' for two days consecutively.

If there are n days, how many ways he can go to office such that he is rewarded a bonus?

E.g.: Consider n = 2. Output: 8
Possible ways: [P, P], [P, A], [P, L], [A, A], [A, L], [A, P], [L, A], [L, P].

8 students were shortlisted for the physical interview held at CUIC.

## Interview:

### Technical Round 1:

This round completely consists of 2 DSA questions which went for nearly an hour. Initially the interviewer asked me to introduce myself. After that the coding session started.

1. Given an array consists of number (integer, fraction, negative, zero), you can delete at most one element in order to maximize the product.

   First the interviewer asked me all possible cases along with output, then asked me to write the code in whitepaper.

2. Given a puzzle, decode that and then write the code in whitepaper.
   Sample:
   Actual String: R E B U S
   Trial String: R O U T E
   Output: + - * - *

   Logic is R is present in both strings at the same position and so + symbol is used, O and T are not in actual string and so – symbol is used, U and E are in actual string but in different positions and so * symbol is used. (Output is with respect to trial string).

   Interviewer asked question about DBMS project that I have done and he gave me some suggestions about it.
   Finally, he asked if I had any questions for him.

### Technical Round 2:

I was asked to give a brief description of the project and after that they went to the coding part.

1. Given an array, we have to find the second largest element.
   First the interview asked me to list out all the approaches that I came up with. After that he asked me about the time complexities of each approach. At last, he asked me to write code for any one of the approaches. He gave me an output and asked for the cases where will I get the specified output.

2. Given a number, we have to find the next largest number using the digits of the given number.

   E.g.: Given number 52394, The next greatest number will be 52439.

   This is similar to the question https://leetcode.com/problems/next-permutation/

   Then he made a slight modification to the problem (to find the next smallest element). He asked me to give approach alone and not to code.

3. Given an array to find whether it is a rotated sorted array or not.

   E.g.: [3,4,5,1,2] is a rotated sorted array (rotated in clockwise direction thrice).

   He asked me to give the approach and to write the code in paper.

**HR ROUND**:

Firstly, the interviewer asked me to introduce myself, my family background and my interests.

Next, he asked me many questions like

1. Where I see myself in next five years?
2. What are the three important things to be followed and three things to be avoided to pursue success in career?
3. What are my strengths and weaknesses?

The round went for nearly 45 minutes and completely full of friendly HR questions.

Out of 8, 2 got shortlisted.

All the best!!!

Selva Rathinam M
Third Year, CSE (Orange tag)
Mobile: 8610120859
Email: selvarathinammurugesan@gmail.com